

1、实验任务

2、设计思路

The diagram illustrates the logic of the LED driver circuit. It consists of three main functional blocks: a '周期信号计数器' (Periodic Signal Counter), an '占空比调节' (Duty Cycle Adjustment) block, and a '比较数值并输出PWM波形' (Compare Value and Output PWM Waveform) block. The '周期信号计数器' block receives two inputs: '50MHz时钟' (50MHz Clock) and '复位' (Reset). It outputs a '计数值' (Count Value) to the '占空比调节' block. The '占空比调节' block also receives the '50MHz时钟' signal and outputs a '占空比' (Duty Cycle) to the '比较数值并输出PWM波形' block. The '比较数值并输出PWM波形' block outputs the 'LED' signal.

3、运行代码

```

1 module breath_led(
2     input    sys_clk    , //时钟信号 50Mhz
3     input    sys_rst_n , //复位信号
4
5     output led          //LED
6 );
7
8 //reg define
9 reg [15:0] period_cnt ; //周期计数器频率: 1khz 周期:1ms 计数值:1ms/20ns=50000
10 reg [15:0] duty_cycle ; //占空比数值
11 reg        inc_dec_flag ; //0 递增 1 递减
12
13 //*****
14 /**                                main code
15 //*****

```

```

16
17 //根据占空比和计数值之间的大小关系来输出 LED
18 assign led = (period_cnt >= duty_cycle) ? 1'b1 : 1'b0;
19
20 //周期计数器
21 always @(posedge sys_clk or negedge sys_rst_n) begin
22     if(!sys_rst_n)
23         period_cnt <= 16'd0;
24     else if(period_cnt == 16'd50000)
25         period_cnt <= 16'd0;
26     else
27         period_cnt <= period_cnt + 1'b1;
28 end
29
30 //在周期计数器的节拍下递增或递减占空比
31 always @(posedge sys_clk or negedge sys_rst_n) begin
32     if(!sys_rst_n) begin
33         duty_cycle <= 16'd0;
34         inc_dec_flag <= 1'b0;
35     end
36     else begin
37         if(period_cnt == 16'd50000) begin //计满 1ms
38             if(inc_dec_flag == 1'b0) begin //占空比递增状态
39                 if(duty_cycle == 16'd50000) //如果占空比已递增至最大
40                     inc_dec_flag <= 1'b1; //则占空比开始递减
41                 else //否则占空比以 25 为单位递增
42                     duty_cycle <= duty_cycle + 16'd25;
43             end
44             else begin //占空比递减状态
45                 if(duty_cycle == 16'd0) //如果占空比已递减至 0
46                     inc_dec_flag <= 1'b0; //则占空比开始递增
47                 else //否则占空比以 25 为单位递减
48                     duty_cycle <= duty_cycle - 16'd25;
49             end
50         end
51     end
52 end
53
54 endmodule

```

第 21-28 行是 1KHz 周期信号的计数器，用于产生 1KHz 的 LED 驱动信号。第 31-52 行的 always 块为占空比设定模块，每次计数完了一个周期，就根据递增/递减标志来对占空比计数值 (duty_cycle) 进行递增/递减 25 个计数值，这个递增或者递减的数值大小可以用来控制呼吸灯的呼吸频率。

如果占空比计数值（duty_cycle）已经递增到了最大，则呼吸灯已经处于最亮的状态，接下来开始递减；反之，如果占空比计数至已经递减到了最小，即 0，则呼吸灯处于熄灭的状态，接下来开始递增；如此循环往复，最终实现了流水灯的效果。

在代码的第 18 行通过组合逻辑把当前的周期计数值和占空比计数值进行比较，来判断 LED 的输出电平。在一个周期内，如果当前的周期计数值小于等于占空比计数值，则 LED 输出高电平，即点亮；如果当前的周期计数值大于占空比计数值，则 LED 输出低电平，即熄灭。

4、引脚约束

```
set_pin_assignment { led[0] } { LOCATION = A4; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { led[1] } { LOCATION = A3; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { led[2] } { LOCATION = C10; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { led[3] } { LOCATION = B12; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { sys_clk } { LOCATION = R7; IOSTANDARD = LVCMOS33;  
PULLTYPE = PULLUP; }  
set_pin_assignment { sys_rst_n } { LOCATION = A2; IOSTANDARD = LVCMOS33;  
PULLTYPE = PULLUP; }
```

5、下载验证

插好下载线，下载程序至核心板，此时可以看到核心板 LED1 发光管由亮到暗，又由暗到亮

